Intelligent Patient Risk Assessment & Agentic Health Support System

Vikrant Yadav (Enrollment No: 2401010504)
Mayank Pillai (Enrollment No: 2401010268)
Prashant Raj (Enrollment No: 2401010340)
Asad Ali (Enrollment No: 2401010108)

April 18, 2026

Abstract

This project implements a comprehensive healthcare analytics system that evolves from predictive modeling to autonomous agentic support. Building upon a two-stage machine learning baseline (Linear Regression and Decision Tree), the system is extended into an Agentic AI application utilizing LangGraph. The system autonomously processes patient risk profiles, retrieves relevant medical knowledge through a Retrieval-Augmented Generation (RAG) pipeline powered by ChromaDB, and generates structured health guidance using the Llama-3.3-70b-versatile model. This methodology bridges the gap between raw statistical risk quantification and actionable, empathetic clinical support.

1 Introduction & Problem Statement

Diabetes mellitus affects over 537 million adults globally, yet traditional diagnostic frameworks often rely on static laboratory thresholds that fail to provide a personalized, continuous "sliding scale" of risk. While Milestone 1 established a robust ML baseline for risk quantification, Milestone 2 addresses the critical need for an autonomous agent capable of reasoning over these predictions.

The core challenge is the integration of high-dimensional clinical data with unstructured medical knowledge to provide structured health reports and conversational follow-up care. The system must autonomously retrieve medical guidelines to ensure recommendations are grounded in verified clinical standards rather than LLM hallucinations.

2 Detailed Data Description

The system utilizes two distinct data sources to power its hybrid architecture:

- **Structured Clinical Data (ML Pipeline):** The system processes the Diabetes Health Indicators Dataset (100,000 records). We pruned 31 original features down to 24 physiological markers, dropping socioeconomic noise such as ethnicity and income levels.
- **Unstructured Medical Guidelines (RAG Pipeline):** To ground the Agentic AI in verified medical science, we utilized the **ICMR Guidelines for Management of Type 2 Diabetes (2018)** and the **WHO Management of Diabetes Mellitus: Standards of Care**. These clinical PDFs were ingested into a **ChromaDB** vector database. They were split into chunks of 1000 characters with a 200-character overlap to preserve semantic context.

3 Exploratory Data Analysis (EDA) Processes

EDA was pivotal in validating the data integrity for the ML baseline:

- **Integrity Checks:** Confirmed zero missing values and zero duplicate records across the dataset.
- **Clinical Insights:** Boxplots identified significant outliers in glucose and cholesterol metrics, while countplots established a strong correlation between `family_history` and diabetes diagnosis.

4 Methodology

The system architecture follows a cascaded design, progressing from statistical prediction to agentic reasoning.

4.1 Agentic Workflow (LangGraph)

The system utilizes **LangGraph** to manage a state-based workflow for report generation. The agent transitions through specific nodes defined in the `HealthAgentState`:

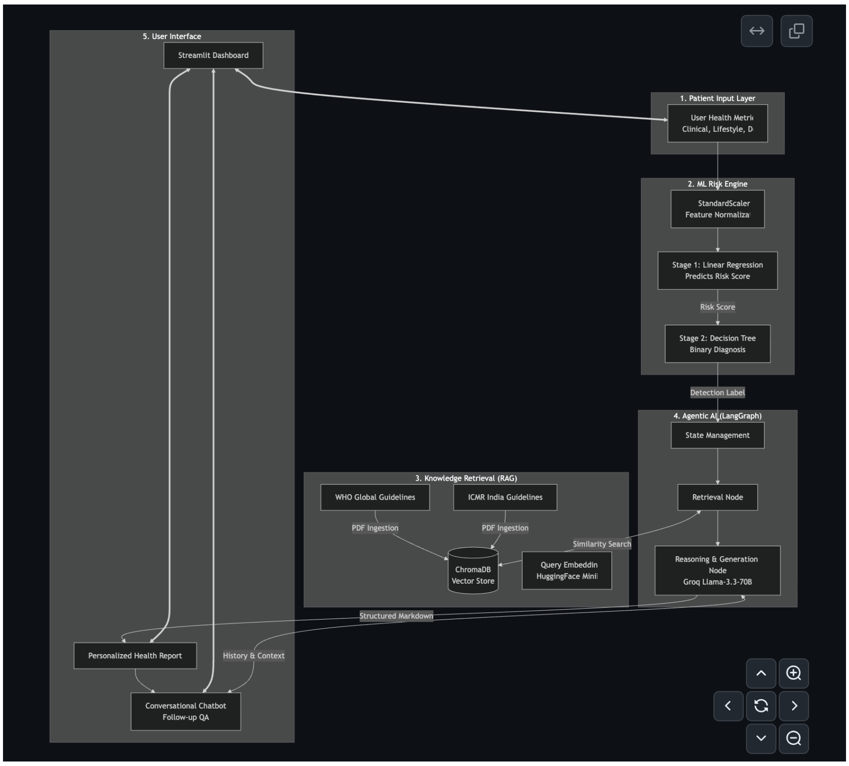

Figure 1: System Architecture: Flow from raw input to ML prediction, followed by LangGraph-driven RAG (ChromaDB) and report generation.

1. **retrieve_guidelines:** The system queries the **ChromaDB** vector store using the patient's risk score and profile to find the top 5 most relevant clinical chunks from the ICMR and WHO guidelines.
2. **generate_report:** The LLM (Llama-3.3-70b) synthesizes the retrieved guidelines with the ML predictions to create a markdown-formatted report.

4.2 Mathematical Formulation

Stage 1: Linear Regression (Risk Scoring)
The continuous risk score $\hat{y}_{risk}$ is calculated as a weighted linear combination of standardized features:

$$\hat{y}_{risk} = \beta_0 + \sum_{i=1}^{n} \beta_i x_i + \epsilon$$

To ensure clinical validity, the output is bounded: $\hat{y}_{clipped} = \min(\max(\hat{y}_{risk}, 0), 100)$.

Stage 2: RAG Retrieval (Cosine Similarity)
Retrieval within ChromaDB is performed by calculating the **Cosine Similarity** between the query embedding (q) and the ICMR/WHO document embeddings (d):

$$\text{similarity} = \frac{q \cdot d}{\|q\| \|d\|}$$

We utilize the `all-MiniLM-L6-v2` model for generating these dense vector embeddings.

5 Optimization Strategies

- **Overfitting Control (ML):** 5-fold KFold cross-validation was implemented on the Linear Regression model.
- **Hallucination Mitigation (Agent):** We implemented strict system prompts and guardrails, ensuring the agent refuses non-medical requests and only makes claims explicitly supported by the retrieved ICMR and WHO guidelines stored in ChromaDB.
- **Hyperparameter Tuning:** The Decision Tree was optimized via GridSearchCV across `max_depth` and `criterion` parameters.

6 Evaluation & Results

ML Pipeline Performance: The Linear Regression model achieved an R^2 of 0.9933. The Decision Tree Classifier achieved 91.94% accuracy and 99.80% precision, producing only 21 false positives across the test set.

Metric	Score
Linear Regression R^2	0.9933
Decision Tree Accuracy	91.94%
Decision Tree Precision	99.80%

Table 1: Performance metrics for the optimized cascaded ML pipeline.

Agentic AI Qualitative Analysis: The agent successfully generates structured reports in Markdown format, including a Risk Summary, Personalized Recommendations, and mandatory Medical Disclaimers. In conversational mode, the ChromaDB RAG pipeline provides highly accurate, context-aware answers to patient follow-up questions, strictly grounded in the established WHO and ICMR guidelines.

7 References

1. **Indian Council of Medical Research (ICMR):** Guidelines for Management of Type 2 Diabetes (2018).
2. **World Health Organization (WHO):** Management of Diabetes Mellitus: Standards of Care and Clinical Practice Guidelines.
3. **Diabetes Dataset:** `https://www.kaggle.com/datasets/mohankrishnathalla/diabetes-health-indicators-dataset`
4. **LangGraph Docs:** `https://langchain-ai.github.io/langgraph/`
5. **Groq Cloud Console (Llama-3):** `https://console.groq.com/docs/models`

8 Team Contribution

- **Vikrant Yadav:** Project Lead; Led ML Preprocessing, EDA, and implementation of the RAG ingestion pipeline for the ICMR/WHO PDFs into ChromaDB (`ingest.py`).
- **Mayank Pillai:** Implemented the Decision Tree Classifier and optimized ML hyperparameter tuning.
- **Prashant Raj:** Developed the Linear Regression model and designed the LangGraph agentic workflow logic.
- **Asad Ali:** Managed LaTeX documentation and led the Streamlit UI integration for the ML and Agentic components.